FFGFT: More than Two States — How a Qubit Encodes

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1 The Real Question: What Lies Between $|0\rangle$ and $|1\rangle$?

Spin-up and spin-down are two points. Two points are unambiguously labelled — one bit of information.

But a **qubit** is not a bit. The real power of the qubit lies not in the fact that it can be $|0\rangle$ *or* $|1\rangle$ — a classical coin can do that too. It lies in the fact that it can be $|0\rangle$ *and* $|1\rangle$ *simultaneously* in a superposition — and in a continuous ratio:

$$|\psi\rangle \;=\; \cos\tfrac{\theta}{2}\,|0\rangle \;+\; e^{i\phi}\sin\tfrac{\theta}{2}\,|1\rangle\,. \tag{1}$$

Two parameters: the mixing angle $\theta \in [0, \pi]$ and the relative phase $\phi \in [0, 2\pi)$. Together they parameterise a **sphere surface** — the **Bloch sphere** (geometric representation of all possible single-qubit states as points on a sphere of unit radius).

On this sphere there are *infinitely many points* — not only the two poles $|0\rangle$ (north pole) and $|1\rangle$ (south pole), but the entire equator, all circles of latitude, every point in between. Every point is a valid, physically distinguishable quantum state.

A qubit does not encode 2 states — it encodes *infinitely many*

The two classical poles $|0\rangle$ and $|1\rangle$ are special cases of a continuous family. What distinguishes a qubit from a classical bit is not the number of discrete states — it is the **continuous geometry** of the state space.
The readout interacts with the system — every measurement couples to the torsion configuration and drives it into the nearest stable extreme state ($|0\rangle$ or $|1\rangle$). The result is deterministic — it depends on the exact initial state which we do not fully know before measurement. The apparent "probability" of standard QM describes this epistemic ignorance, not fundamental randomness.
The states themselves are discrete. The continuity lies in the phase space *before* measurement — the readout always delivers one of the two stable poles.

2 The Cylindrical Phase Space — FFGFT Geometry of the Qubit

The Bloch sphere is the standard representation. In FFGFT (Doc. 034) there is a physically more direct alternative: the **cylindrical phase space**.

A qubit state is described by three geometric coordinates:

$$(z,\ r,\ \theta), \tag{2}$$

where:

- $z \in [-1, +1]$: the "height" along the cylinder axis — describes the proportion of the torsion configuration in the direction of $|0\rangle$ vs. $|1\rangle$. ($z = +1$: pure torsion type $|0\rangle$; $z = -1$: pure torsion type $|1\rangle$)
- $r \in [0, 1]$: the radius in the cross-section — indicates the strength of the intermediate torsion, i.e. how far the state is from both poles. ($r = 0$: stable pole; $r = 1$: maximum intermediate torsion)

- $\theta \in [0, 2\pi)$: the azimuthal angle — the **relative phase** between the two basis components.

The relation to the Bloch sphere is direct: $z = \cos\theta_{\text{Bloch}}$ and $r = |\sin\theta_{\text{Bloch}}|$. But the cylinder is not a geometric decoration — it makes gate operations into *geometric transformations* in real space:

Gate	Hilbert space (matrix)	FFGFT cylinder (transformation)
Hadamard H	$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$	Swap $z \leftrightarrow r$, rotate phase by $\pi/2$ — moves pole to equator
Phase Z	$\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$	Add π to phase θ — pure rotation about cylinder axis
Bit-flip X	$\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$	2D rotation of (z, r) by angle $\alpha = \pi \cdot K_{\text{frak}}$ — fractal damping built in

FFGFT: gates as torsion rotations

In the cylindrical FFGFT formalism every gate operation is a geometric transformation of the torsion vector. The Hadamard gate does not rotate a matrix — it tilts the torsion configuration from the longitudinal axis to the transverse plane. The Z gate rotates the phase θ — it rotates the torsion winding about its own axis.

The factor K_{frak} in the X gate is not a correction factor added afterwards — it follows directly from the fractal spacetime geometry ($D_f = 2.999867$) and means: every qubit rotation is slightly smaller than π, because spacetime itself minimally deviates from flat 3D geometry. This is a testable prediction: all π gates should systematically deviate by $\Delta\alpha \approx 0.013\,\%$ from the ideal value.

3 Quantum Registers — Exponential Growth through Combination

A single qubit has infinitely many intermediate states in the torsion phase space, but always delivers one of the two stable poles upon readout — the readout itself is the interaction that drives the state to the next stable extreme. The real power comes through **combination**.

3.1 Product States: Additive Classicality

Two independent qubits have together $2 \times 2 = 4$ basis states: $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$.

The general state *without entanglement* (product state) is:

$$|\psi_A\rangle \otimes |\psi_B\rangle = (\alpha_A |0\rangle + \beta_A |1\rangle) \otimes (\alpha_B |0\rangle + \beta_B |1\rangle). \tag{3}$$

This is only two separate Bloch spheres — no gain over two independent qubits. The state space is *additive*: $2 + 2 = 4$ parameters.

3.2 Entangled States: Multiplicative Growth

Once the two qubits are *entangled*, the joint state can no longer be written as a product of two individual states. The general two-qubit state requires 4 complex amplitudes (minus normalisation and global phase: $2 \times 4 - 2 = 6$ real parameters):

$$|\Psi\rangle = \alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \delta |11\rangle . \tag{4}$$

This is **more** than two independent qubits — entangled qubits carry joint information that is contained in neither individual qubit.

For n qubits the Hilbert space grows exponentially:

$$\boxed{d(n) = 2^n} \tag{5}$$

Real parameters in the general state: $2 \cdot 2^n - 2$.

n qubits	Basis states	Real parameters	Classical equivalent
1	2	2	1 bit
2	4	6	2 bits
3	8	14	3 bits
10	1,024	2,046	10 bits
50	10^{15}	2×10^{15}	50 bits
300	10^{90}	2×10^{90}	$\approx$ number of atoms in the universe

This is the actual quantum advantage: Not a single qubit is powerful — but the collective growth of an *entangled* register.

3.3 Bell States — Maximum Entanglement in Two Qubits

The four **Bell states** are the maximally entangled two-qubit states — each a complete orthonormal basis of the entangled subspace:

$$|\Phi^+\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \tag{6}$$
$$|\Phi^-\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \tag{7}$$
$$|\Psi^+\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \tag{8}$$
$$|\Psi^-\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \tag{9}$$

Each of these states has the property: if qubit A is measured, the result of qubit B is immediately determined — regardless of how far away.

In FFGFT (Doc. 023, 035):

$$E^{\text{FFGFT}}(a, b) \;=\; -\cos(a - b) \cdot \left(1 - \xi\, \frac{(\Delta\theta)^2/\pi^2}{D_f}\right), \tag{10}$$

the ξ damping reduces the CHSH violation from $2\sqrt{2} \approx 2.8284$ to ≈ 2.8275 — a difference of $\sim 10^{-4}$, which explains non-locality locally without eliminating it.

FFGFT: Bell states as joint torsion nodes

In the FFGFT picture a Bell state is not an action at a distance between two qubits — it is a **topologically connected torsion configuration** across two resonators. The two resonators share a common torsion node whose resolution simultaneously determines both ends at every local measurement.

This is not a signal propagating faster than light — it is a common geometric state established at preparation. The local readout interaction at one end deterministically drives the local torsion into a stable pole — and thereby necessarily also determines the state of the other end, because both share the same torsion node. The "global consequence" is not action at a distance, but the geometric coherence of the common node.

The ξ damping (FFGFT correction to Bell correlations) is in this picture the finite extension of the torsion node: the node is not point-like, but has a characteristic length $\sim L_\xi = \xi \cdot \ell_P \cdot (1/\xi)^N$ at the ξ scale level $N \approx 8$.

IBM Quantum hardware validation: Bell states, June 2025

The FFGFT predictions for Bell states were experimentally verified on genuine quantum hardware:

Hardware: IBM Brisbane & IBM Sherbrooke (127 qubits each, Heavy-Hex lattice), access via IBM Quantum Platform / Qiskit Runtime.

Result:

Quantity	Value
Bell state fidelity	$97.17\,\%$
FFGFT algorithm vs. QM	Agreement within measurement precision
Repetition variance	0.0001–0.001 (excellent)
Deviation $P(00)$, $P(11)$	1–$5\,\%$ (hardware noise)

Significance in FFGFT context: The FFGFT amplitudes $E_i(x,t)$ with ξ coupling ($\xi = 1.038 \times 10^{-5}$, Higgs-derived) deliver algorithmically equivalent predictions to standard QM — with improved repeatability (lower variance). The ξ deviation from the ideal Bell correlation (ΔCHSH $\approx 10^{-5}$) lies below the current hardware noise threshold and could therefore not yet be directly isolated and measured. This is a precise experimental task for future high-precision Bell tests.

4 Three Ways to More States — a Comparison

Strategy	More states through	Growth	FFGFT meaning
Superposition (1 qubit)	Continuous superposition of $\|0\rangle$ and $\|1\rangle$	∞, but 1 bit at measurement	Intermediate torsion on the Bloch sphere
Register (n qubits)	Entanglement of multiple qubits	2^n (exponential)	2^n-dimensional torsion product space; entangled = topologically connected nodes
Qudit ($d > 2$, 1 object)	Higher energy levels, orbital states, exciton number	d (linear) per object, d^n for n qudits	d stable torsion configurations in one resonator

No strategy dominates in all areas

Register entanglement scales most strongly (2^n), but requires n physical objects and error-corrected entanglement operations between them.
Qudit coding gains information *per object* ($\log_2 d > 1$), but the exponential growth is slower.
Superposition is always present — it is the basis for quantum interference, the actual driving force of quantum algorithms.

5 Quantum Interference — Why Superposition is Useful

Superposition alone would be a mere overlay — the readout would always deliver one of the two stable poles, deterministically dependent on the exact phase state.

What makes superposition into a *resource* is **interference**: amplitudes can constructively (amplify) or destructively (cancel) overlap — depending on phase ϕ.

This is the principle behind all useful quantum algorithms:

1. Prepare the register in a uniform intermediate torsion over all inputs (with the Hadamard gate).
2. Apply a function that increases the phases of the correct answers and decreases those of the wrong ones.
3. Let interference drive the wrong answers into unstable torsion configurations and the correct one into a stable one.
4. The readout drives the system deterministically into the nearest stable pole — which after interference corresponds with overwhelming frequency to the sought answer.

Without interference: Monte Carlo (classical guessing). With interference: Grover ($\sqrt{N}$ instead of N steps), Shor (factorisation in polynomial time).

FFGFT: interference as torsion phase superposition

In FFGFT the phase ϕ is not an abstract complex number — it is the rotation position of the intermediate torsion configuration (Doc. 174).

Constructive interference: two torsion phases coincide — they amplify each other into a geometrically more stable configuration. Destructive interference: two opposite torsion phases cancel each other — the resulting configuration has no stable geometric anchor and is not addressed at the readout interaction.

A quantum algorithm in FFGFT language: a sequence of torsion rotations that aligns the phase configuration of the register so that the sought answer ends in a stable torsion configuration — all others in geometrically unstable ones. The readout is then not a random selection, but a deterministic reaction of the resonator to the readout interaction.

6 QND Measurement — Reading Without Destroying

6.1 What QND Means

A QND measurement reads an observable $\hat{A}$ without changing the corresponding eigenstate. The formal condition:

$$[\hat{A},\ \hat{H}_{\text{measurement}}] = 0. \tag{11}$$

The measurement operator $\hat{H}_{\text{measurement}}$ commutes with the measured observable — it does not change the eigenvalue, because it respects the same eigenspace.

Practically this means: the same state can be **repeatedly measured** and delivers the same result each time — because the measurement does not touch the state itself.

The most important example from Doc. 174: **Faraday/Kerr rotation**. A linearly polarised laser beam passes the quantum dot without being absorbed — it rotates its plane of polarisation by $\theta_F \propto S_z$. No photon is absorbed. No energy transfer into the spin mode. The spin state before and after measurement is identical.

6.2 QND as Empirical Proof of Determinism

The QND measurement is from the FFGFT perspective far more than a technically useful property — it is an **empirical proof** for the reality of stable torsion configurations.

The argument is direct:

1. Faraday rotation measures S_z without changing the spin state.
2. Repeated measurements always yield the same result.
3. Therefore the state had this value *before every measurement* — definitely, not as an abstract superposition that only “becomes reality” through measurement.
4. The state was real before we looked.

FFGFT: QND measurement and stable torsion configuration

In FFGFT, Faraday rotation is state-preserving because a stable torsion configuration (spin-up = topologically fixed winding in one direction) cannot be driven out of its state through a dispersive, non-resonant coupling.

The photon "sees" the torsion — it rotates its polarisation accordingly — but it transfers no energy into the torsion mode, because it is not resonant. The geometric state is invariant under this coupling.

The philosophical consequence is clear: If a state can be repeatedly measured without disturbance, then it was not "created" by the measurement — it was there before. The standard QM interpretation ("the state exists only through measurement") is not compatible with QND measurements.

In FFGFT: the torsion configuration is a geometric reality — stable, definite, pre-existent. The measurement reads it off. It does not create it.

6.3 Potential Landscape: Stable Minima, Transitions and Readout Strength

FFGFT: torsion potential, transitions and readout strength

The complete picture of qubit states is a **potential landscape with two stable minima** and a saddle between them.
The readout strength decides what happens:

Initial state	**Weak coupling**	**Strong coupling**
Stable minimum ($\|\uparrow\rangle$ or $\|\downarrow\rangle$)	Coupling does not lift state from minimum — **QND**: state remains, result reproducible	Coupling can lift state from minimum — invasive, state changed
Slope / saddle (superposition)	Even weak coupling is sufficient to drive the state — it "rolls" into the next minimum. **Which one**: determined by exact phase ϕ (epistemically unknown, deterministic)	Same, just faster — state jumps immediately into next minimum

The decisive insight: There is no principled boundary between "QND-possible" and "QND-impossible" — there are only **more stable and less stable states** in a continuous potential landscape.
A state in the deep minimum is stable against the readout coupling — the minimum must be overcome. A state on the slope is unstable — any coupling, however weak, is sufficient to drive it into the next minimum.
And the "jump" to the final position is not random — it is determined deterministically by the phase of the initial state on the slope: a state near $|\uparrow\rangle$ lands in $|\uparrow\rangle$, a state near $|\downarrow\rangle$ lands in $|\downarrow\rangle$, a state on the saddle ($\theta = \pi/2$) lands depending on the fine phase ϕ.
The apparent "randomness" of standard QM is the epistemic ignorance of this phase — not an ontological randomness of the system.
The picture applies to all d levels of a qudit: not two minima, but d minima — with $d-1$ saddle points between them.

QND and superposition — no contradiction

A stable eigenstate ($|\uparrow\rangle$ or $|\downarrow\rangle$): sits in the deep minimum — weak coupling reads without driving. QND possible. Result reproducible.
An intermediate torsion ($\alpha\,|\uparrow\rangle + \beta\,|\downarrow\rangle$): sits on the slope — any coupling drives into the next minimum. Deterministic, not random. The phase ϕ determines which minimum is reached.
The QND measurement *proves* the reality of stable poles. The disturbing measurement *proves* the instability of intermediate torsions — and the determinism structure of the potential landscape. Together: completely deterministic, geometrically founded picture.

7 Two Levels of the ξ-Hierarchy: Spin Qubit and Photonic Resonator

7.1 Spin Qubit: Minimal Resonator at Level $N \approx 8$

A spin qubit is the absolute minimum of a quantum resonator (Doc. 173): a single electron, a single energy level, two permitted orientations.

The energy scale comes from **quantum confinement** at ξ level $N \approx 8$ (nanometre scale). The spacing between the two spin states in a typical magnetic field ($B \approx 1\,\text{T}$) lies at:

$$\Delta E_Z = g^* \mu_B B \approx 0.1\text{–}1\,\text{meV}. \tag{12}$$

The thermal energy at room temperature is:

$$k_B T(300\,\text{K}) = 26\,\text{meV}. \tag{13}$$

So $\Delta E_Z \ll k_B T$ — thermal noise constantly knocks the system out of the minimum. Low temperatures ($T < 1\,\text{K}$, $k_B T < 0.1\,\text{meV}$) are not optional, but mandatory — not to maintain superposition, but simply to *stay* in one of the two minima.

7.2 Photonic Resonator: Collective Geometry at Level $N \approx 9$–10

A waveguide of lithium niobate ($LiNbO_3$), a Mach-Zehnder interferometer, a ring resonator — these are resonators made of 10^{18} to 10^{23} atoms in perfect lattice order.

The decisive difference from the spin qubit: **the resonator is not a single particle — it is the macroscopic geometry of the entire crystal.**

The energy scale of an optical photon:

$$E_{\text{photon}} \approx 1\text{–}2\,\text{eV} \quad (780\text{–}1550\,\text{nm}), \tag{14}$$

so:

$$\frac{E_{\text{photon}}}{k_B T(300\,\text{K})} \approx \frac{1\,\text{eV}}{26\,\text{meV}} \approx 40. \tag{15}$$

Thermal noise cannot spontaneously create optical photons — the system sits far below the thermal threshold for optical excitations. Condition 2 from Doc. 173 ($\Delta E \gg k_B T$) is satisfied by a factor of 40.

Why photonics works at room temperature

Not because the photonic system is more robustly built — but because it operates at a higher ξ level of the energy hierarchy.

The **geometric robustness** (Condition 1) is extraordinarily well satisfied by the macroscopic single crystal with $\sim 10^{23}$ lattice sites in perfect order — much more robust than a single quantum dot. Individual phonons (thermal lattice vibrations) barely disturb the collective mode.

The **thermal robustness** (Condition 2) is automatically given by the eV energy scale — thermal photons in the infrared have $\sim 25\,$meV, optical photons have $\sim 1\,$eV: four orders of magnitude safety margin.

7.3 Comparison of the Two Levels in the ξ-Hierarchy

	Spin qubit	**Photonic resonator**
ξ **level**	$N \approx 8$ (nm scale)	$N \approx 9$–10 (μm–mm scale)
Resonator	1 electron in nm cage	10^{18}–10^{23} atoms as collective crystal
Energy scale	$\Delta E \approx 0.1$–1 meV	$E_{\text{photon}} \approx 1$–2 eV
$\Delta E/k_B T$ **(300 K)**	≈ 0.004–0.04 (too small!)	≈ 40 (robust)
Cooling needed?	Yes, $T < 1$ K mandatory	No, room temperature
Analogue states	Fragile: $T_2 \sim$ ns at 300 K	Stable: phase thermally unreachable
Discreteness from	Quantum confinement (1 particle)	Macroscopic mode geometry
Typical application	Digital qubit, quantum gate	Analogue signal processing, photonic computing, 6G

FFGFT: two levels of the ξ hierarchy — same physics, different scales

Spin qubit and photonic resonator are in FFGFT not fundamentally different systems — they are resonators at different levels of the ξ scaling ladder.

At level $N \approx 8$: the confinement energy minimum lies at meV — not thermally robust at 300 K. At level $N \approx 9$–10: the collective mode minimum lies at eV — completely thermally robust.

The ξ parameter generates a universal energy hierarchy: each level carries the energy by a factor of $1/\xi \approx 7500$ higher than the previous. From meV (level 8) to eV (level 9–10): the factor is $\sim 10^3$–10^4 — consistent with ξ scaling.

This explains not only photonics at room temperature — it explains why biological systems use photonic mechanisms (photosynthesis, bird navigation via light) and not spin qubit systems for quantum effects: nature works at the thermally robust level.

8 ξ-Level Separation as Stability Principle

8.1 The Coupling Formula Between Levels

Between different ξ levels, the coupling strength decreases exponentially with the level difference ΔN:

$$\Gamma_{\Delta N} \propto \xi^{\Delta N}. \tag{16}$$

For $\Delta N = 1$: $\xi^1 \approx 1.333 \times 10^{-4}$ (weak but non-zero). For $\Delta N = 2$: $\xi^2 \approx 1.78 \times 10^{-8}$ (extremely weak). For $\Delta N = 4$: $\xi^4 \approx 3.2 \times 10^{-16}$ (practically non-existent).

8.2 Fractal Self-Similarity *Because of* the Damping

The fractal hierarchy of nature is not stable *despite* ξ damping — it is stable *because of* it.

Were ξ not small, all levels would strongly couple to each other. Every change at one level would immediately restructure all other levels. There would be no stable atoms, no stable molecules, no stable cells — only a single chaotic coupled system of all scales simultaneously.

Because $\xi \ll 1$, the levels can exist **quasi-independently**: each level has its own stable internal dynamics, barely disturbed by the other levels.

FFGFT: ξ damping as origin of the natural hierarchy

The existence of separate, stable structural levels in nature — quarks, nucleons, atoms, molecules, cells, organisms, planets, galaxies — is not a random property of matter.
It follows directly from ξ level separation: each level is practically invisible to the next-but-one. The coupling falls exponentially with level distance. This is the mechanism that makes the fractal self-similarity of nature *stable*.

9 Qudit Register — the Connection to the ξ-Hierarchy

From Doc. 174: a quantum dot of dimension d encodes $\log_2 d$ bits. What happens when one entangles n such qudits?

The total state space has dimension d^n. For $d = 4$ (spin-orbital ququart) and n qudits:

$$d^n = 4^n = (2^2)^n = 2^{2n}. \tag{17}$$

This is equivalent to $2n$ qubits in a register of size n — double the information density per object.

Qudit density at ξ level $N \approx 8$

A quantum dot with $R = 3$ nm at 4 K has at least 4 thermally stable states (spin-orbital ququart, $d = 4$). A register of $n = 10$ such qudits has dimension $4^{10} = 1,048,576$ — the same as 20 classical qubits, but only 10 physical objects at ξ level $N \approx 8$.

10 The FFGFT Topology Compiler — Geometry of Entanglement

A practical problem with entangled registers: entanglement gates between non-adjacent qubits are expensive — they require many SWAP operations (which accumulate errors and cost time).

FFGFT (Doc. 034) offers a geometric solution: the **FFGFT topology compiler**.

Standard compilers minimise the number of SWAPs. The FFGFT compiler instead minimises the cumulative **fractal path length** of all entanglement operations. Since the fractal damping term exp$(-\xi \cdot \ell / D_f)$ is distance-dependent (larger physical distance ℓ means stronger ξ damping of coherence), critically interacting qubits must be *physically placed closer* to each other.

This leads to a new chip architecture: not laid out according to logical circuit optimisation, but according to **geometric coherence maximisation**.

Connection to the adaptive hybrid architecture

The adaptive hybrid architecture (Doc. 173 conclusion) selects resonators according to task: quantum dots for local synaptic operations, Ising machines for global optimisation.

The FFGFT topology compiler is the linking element: it decides which qubits/qudits must physically sit close together (strong entanglement) and which may remain further apart (weakly, tolerantly damped entanglement).

The learning AI on this hardware does not only learn *what* is to be computed — it also learns *which geometric arrangement* executes the computation most coherently. Not through programmed topology specifications, but through geometrically informed learning in the FFGFT compiler.

11 Comparison: Classical Bit, Qubit, Qudit, Register

	Class. bit	**Qubit**	**Qudit (d)**	**n qubits**
State space	2 points	Bloch sphere (∞)	$\mathbb{CP}^{d-1}$ (∞)	$\mathbb{CP}^{2^n-1}$ (∞)
Readout result	1 bit (deterministic)	1 bit (deterministic, depending on phase state)	$\log_2 d$ bits (deterministic)	n bits
Parallel processing	none	Interference over 2 paths	Interference over d paths	Interference over 2^n paths
Resource for advantage	—	Phase ϕ	$d-1$ phases	Entanglement + phases
FFGFT picture	stable/unstable torsion	Torsion on Bloch sphere	d stable torsion levels	Torsion network, topologically connected

12 Open Connections — Next Steps

Open questions for further chapters

1. Fractal gate corrections measurable?
The FFGFT compiler says: all π pulses deviate by $\Delta\alpha \approx K_{\text{frak}} - 1 \approx 0.013\,\%$ from the ideal π. Current superconducting qubit systems achieve gate fidelity $> 99.9\,\%$. That is just at the boundary to see this systematic deviation. Is there a systematic bias in the π pulse length in current calibration data?

2. Bell states and ξ scale level:
The ξ damping in Bell correlations (ΔCHSH $\approx 10^{-4}$) depends on $N \approx 8$. For Bell tests with *macroscopic* objects (atoms, molecules) the scale level N would increase — and with it the damping. Is there an N-dependent suppression of Bell violations for different system sizes?

3. Qudit entanglement and torus topology:
Two entangled qudits ($d = 4$) correspond in FFGFT to two topologically connected spin-orbital torsion configurations. What is the geometric description of this connection in the toroidal field? Is there a conserved quantity (topological invariant) that quantifies the entanglement depth?

4. Quantum error correction and ξ granularity:
The surface code needs ~ 1000 physical qubits per logical qubit. In a qudit-based register with $d = 4$ one needs only half as many physical objects for the same logical information. How does the error correction threshold change in the FFGFT framework when the natural noise floor $\xi \approx 1.333 \times 10^{-4}$ applies as the lower error bound?

13 Summary

Conclusion: More than Two States

A single qubit has infinitely many intermediate states in the torsion phase space — but the states themselves are **discrete**: the readout always delivers one of the two stable poles, deterministically determined by the exact phase state at the moment of readout. The apparent "probability" of standard QM is epistemic ignorance of this state — not fundamental randomness of the system.

The readout interacts with the system — but not always destructively. Faraday/Kerr rotation (Doc. 174) is a QND measurement: it reads the state without changing it, because the coupling commutes with the spin operator. That the same state can be repeatedly measured is the empirical proof that it *existed before* measurement — definitively and really. The result remains deterministic in every case.

The actual quantum advantage comes through three routes:

1. Superposition: Continuous intermediate torsion with phase ϕ enables interference — the geometric foundation of all useful quantum algorithms. Interference aligns the register so that the sought answer ends in a stable torsion configuration; the readout follows that deterministically.

2. Register entanglement: n qubits span a 2^n-dimensional torsion space. Entanglement means: the torsion nodes of the qubits are topologically connected — reading one node necessarily also determines the others. Bell correlations are locally explained by ξ damping without eliminating non-locality.

3. Qudit coding: A resonator with $d > 2$ stable torsion levels delivers $\log_2 d > 1$ bits per object at readout. Spin-orbital ququarts ($d = 4$) at ξ level $N \approx 8$ double the information density of a register.

In FFGFT all three strategies are expressions of the same fundamental principle: stable torsion configurations in geometrically precise resonators — discrete from inside, analoguely controllable from outside, deterministic at readout.

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